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Correlation between HLA-DQB1*06:02 and narcolepsy with and without cataplexy: approving a safe and sensitive genetic test in four major ethnic groups. A Systematic Meta-analysis.

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ABSTRACT

Study Objectives: we performed a meta-analysis to assess the usefulness of HLA testing for Narcolepsy diagnosis in four major ethnical groups: Asians, Afro-Americans, Amerindians, Caucasians.

Methods: PubMed, EMBASE, Web of Science, Scopus and Cochrane databases were searched for articles in English and French published before October 2017 on HLA class II alleles in Narcolepsy. We included case-control studies, cross-sectional and retrospective cohort studies with patients diagnosed following the International classifications of sleep disorders (1990-2012) and ethnically matched controls. Following PRISMA guidelines, two investigators independently extracted data according to the inclusion criteria listed in PROSPERO CRD42017058677. A third researcher was consulted for discrepancies. We extracted and pooled adjusted OR using random-effect models. We verified the strength of the association between HLA-DQB1*06:02 and the worldwide distribution of Narcolepsy type 1 (NT1) and type 2 (NT2); we pooled the OR measuring the association between HLA-DQB1*06:02 and NT1, NT2 and hypersomniacs.

Results: We identified 511 titles; 12 case-control studies were included, for a total of 2077 NT1 patients, 235 NT2 patients, 161 hypersomniacs and 7802 controls. In the population-stratified analysis, HLA-DQB1*06:02 conferred an increased risk for NT1 (OR:24.1, IC:14.6-39.5, $p<0.001$), and NT2 (OR:3.9; IC:2.2-6.8, $p<0.001$). For NT1 the pooled estimated positive Likelihood Ratio (LR+) was 5.94 (IC:3.71-9.51) and the negative Likelihood Ratio (LR-) was 0.23 (IC:0.16-0.33); for NT2 LR+ was 3.35 (IC:2.08-5.38) and LR- 0.72 (IC:0.63-0.81); for hypersomniacs LR+ was 1.436 (IC 0.668- 3.089) and LR- 0.903 (IC 0.714- 1.142).

Conclusions: Our data support the preponderant role of HLA-DQB1*06:02 in susceptibility to NT1/NT2 across all ethnicities. HLA-DQB1*06:02 negativity should make clinicians cautious in excluding other diagnoses.

Keywords: Narcolepsy; Cataplexy; Human Leukocyte Antigens, DQB1*06:02; Autoimmunity.

1. INTRODUCTION

Narcolepsy is a chronic hypersomnia disorder highlighted by sleep attacks and cataplexy as core symptoms.¹ ICSD-2nd divided narcolepsy into two subtypes based on the presence or absence of cataplexy. ICSD-3rd criteria, based on the concept that the absence of hypocretin is considered a fundamental indicator as well as moving diagnostic criteria towards molecular biomarkers, reclassified narcolepsy into narcolepsy type 1 (NT1) and type 2 (NT2). NT1 is associated with cataplexy and (or) low levels of hypocretin, whereas NT2 is associated with the absence of cataplexy and normal (or unknown) levels of hypocretin. Indeed, from an etiopathogenic point of view, narcolepsy with cataplexy is caused by hypocretin deficiency due to selective loss of hypocretin-producing neurons in the hypothalamus.²⁻⁴

Epidemiological studies have shown that Narcolepsy with cataplexy affects 20-50/100000 individuals in the European and USA populations. Worldwide, the highest prevalence is among Japanese (160/100000 individuals), then among Chinese (30/100000 individuals) and South Koreans (15/100000 individuals), while the lowest prevalence in the world is among Israelis (0.23/100000 individuals).⁵ NT1 and NT2 susceptibility is deemed to be complex, including both environmental influences and polygenetic factors.⁶

Since the first International Classification of Sleep Disorders (ICSD) diagnostic criteria were set, Human Leukocyte Antigen (HLA) typing has been suggested as a useful test to detect high-risk subjects in families of narcoleptic patients. Subsequently, ICSD-2nd suggested the HLA-DQB1*06:02 allele was a marker of a genetic predisposition to disease development. Following ICSD-3rd criteria, NT1 may be diagnosed by either cerebrospinal fluid hypocretin-1 deficiency or a mean latency of ≤ 8 minutes and two or more SOREMPs on a multiple sleep latency test (MSLT) (or one SOREMP on a polysomnogram and one or more on MSLTs) and clear-cut cataplexy.²⁻⁴ Criteria for NT2 diagnosis state that cataplexy must be absent and cerebrospinal fluid hypocretin-1 levels must be normal or unknown; it requires a mean sleep latency of ≤ 8 minutes and two or more SOREMPs on an MSLT (or one SOREMP on a polysomnogram and one or more on MSLTs).⁷ About ten percent of patients with cataplexy and normal levels of hypocretin in CSF have been reported; these patients are classified as NT1 according to ICSD3rd.^{8, 9} The invasiveness of lumbar puncture to draw

CSF for measuring hypocretin compels us to find an easy-to-handle marker for narcoleptic patients with cataplexy.

Unlike genetics, immunogenetics markers like HLA alleles are not used alone to make a direct diagnosis, as is the case for genomic mutations with Mendelian diseases. Specific HLA alleles suggest only a predisposition to developing a particular immune-related disease. Many HLA alleles linked to immune-disorders are present with medium-high frequencies worldwide. For example, HLA-B*27, which is historically associated with ankylosing spondylitis, shows a global frequency of 10% from populations all over the world (allele frequencies.net). Therefore, HLA markers indicate a predisposition to develop a disease concomitant with specific clinical signs.¹⁰

In 2002, Okun and colleagues proposed HLA typing as an alternative to invasive methods to validate cataplexy diagnosis.¹¹ In 2008, Bourgin and colleagues recommended measuring hypocretin levels only when dealing with cataplexy of psychogenic origin, in conjunction with negative multiple sleep latency tests and after HLA-DQB1*06:02 positivity.² Knudsen and co-authors suggested that HLA typing should precede CSF hypocretin measurements because hypocretin deficiency is rare in HLA-DQB1*06:02 negative patients.⁸

Starting in 2009, several epidemiological studies showed an increased incidence of NT1 after H1N1 vaccination in individuals carrying the HLA-DQB1*06:02 allele. The most endorsed model is a molecular mimicry mechanism with the H1N1 viral peptides: a specific H1N1 peptide is presented by the HLA-DQB1*06:02 molecules to CD4 T lymphocytes by antigen-presenting cells, the activated CD4 and CD8 cells recognize the H1N1 viral peptide and infiltrate the hypothalamus where the destruction of hypocretin neurons is mediated by inflammatory cytokines.³ Based on these findings, NT1 is one of the first immune-mediated diseases for which the immune-pathogenetic process has revealed the link between HLA and disease development.

We performed a meta-analysis of the association between HLA-DBQ1 alleles and the susceptibility to narcolepsy (both NT1 and NT2), for the purpose of providing additional insights into the genetic predisposition to narcolepsy.

As a first step we performed a meta-analysis to quantitatively evaluate the association between HLA-DQB1 alleles and susceptibility to narcolepsy, bearing in mind the “old” (Narcolepsy with and without cataplexy) and the “new” (Narcolepsy type 1 with low CSF Htrcr-1 level and Narcolepsy type 2 with normal CSF Htrcr-1 levels) narcolepsy definitions and coding. We also added an analysis of the association between hypersomnia and HLA-DQB1*06:02. We then included different ethnic groups in the meta-analysis to verify the strength of the association between the HLA-DQB1*06:02 allele and the worldwide distribution of Narcolepsy type 1 and type 2.

2. METHODS

This work was prepared according to PRISMA guidelines.¹² We drew up a protocol including review questions, selection and eligibility criteria, primary and secondary endpoints, search strategy, methods for data extraction, methods for assessing study quality and risk of bias, strategy for data synthesis, and statistical methodology. On 16 March 2017, the protocol entitled “Association between HLA class II (DQB1, DQA1, and DRB1) polymorphisms and Narcolepsy: a meta-analysis” was published in the PROSPERO International Prospective Register of Systematic Reviews (PROSPERO CRD42017058677); from that date forward it is available from <http://www.crd.york.ac.uk/PROSPERO/>.

2.1 Search Strategy

We performed a systematic search of PubMed, EMBASE, Web of Science, Cochrane databases, and Scopus retrieving all publications (case-control study, cross-sectional and retrospective cohort study) on the association between HLA class II polymorphism (DQB1, DQA1, and DRB1) and Narcoleptic patients (>18 years). We searched all English and French articles published in up to October 2017. An expert librarian performed the search using the following MeSh terms: ("Narcolepsy") AND ("Histocompatibility Antigens Class II") AND (“allele” OR “gene” OR “variant” OR “type”).

Selection criteria were: HLA class II molecular typing (DQB1, DQA1 and DRB1); Narcolepsy diagnosed following the clinical criteria of 1) American Sleep Disorder Association, Diagnostic Classification Steering Committee, Thorpy MJ, Chairman (1990) International Classification of Sleep Disorders: Diagnostic and

Coding Manual. Rochester, Minnesota: ASDA; 2) The International Classification of Sleep Disorders – Revised (1997); 3) International Classification of Sleep Disorders: Diagnostic and Coding Manual. 2. Westchester: American Academy of Sleep Medicine (2005); 4) Criteria from International Classification of Sleep Disorders 3 ed.^{1, 7}

Two investigators independently performed data extraction according to our selection criteria. The third participant was consulted for discussion to reach an agreement concerning discrepancies. The following items were extracted from each study: first author's last name, publication date, country of origin, numbers of cases and controls, HLA typing method. Figure 1 shows the complete flow diagram following the PRISMA statement.¹²

2.2 Selection of Published Studies and Quality Assessment

Following a quality assessment tool for genetic data [Quality Assessment of Genetic Studies in Systematic Reviews (QUAGENS)] proposed by our multidisciplinary panel (statisticians, clinical epidemiologists, geneticists, clinicians, meta-analysts), three pairs of reviewers (one for the clinical criteria, one for laboratory issues, one for methodological tools) working independently and with adequate reliability, verified the following aspects:

1. Clinical data: the presence of spectrum disease biases, the possible enrollment of incident or prevalent cases, the inclusion of controls not selected from the same source population as the case-subjects, the occurrence of differential participation in cases and controls.
2. Laboratory issues: the misclassification of genotypes or serotypes (including the types and quality of samples, the timing of sample collection, and the method used for typing), the actual laboratory staff blinded to outcome, the mention of quality controls.
3. Methodological features: the possible population stratification, the presence of multiple testing and pre-study odds of true finding (it would be useful to interpret the results in the context of how many polymorphisms have been studied), the assessment of HW equilibrium in controls.

Each question was answered as “yes,” “no” or “unclear.”¹³

2.3 Data Extraction

After a critical reading of the articles, two investigators independently performed data extraction according to the inclusion criteria listed above. The third researcher was consulted for discussion to reach an agreement concerning discrepancies.

2.4 Data synthesis

STATA 14.2 (StataCorp USA) and METADISC 1.4 were used for statistical analysis to perform the meta-analysis. Heterogeneity was checked by the Chi2 test and the I-squared statistic. The criteria for identification of heterogeneity were a P value less than 0.10 for the Chi-squared test and an I-squared statistic greater than 50%. When there was no statistical evidence for heterogeneity in effect sizes, we used the fixed-effect model to meta-analyze Odd Ratios (ORs) or Relative Risks (RRs) in probands; when significant heterogeneity was identified, we used the random-effects model and explored sources of significant heterogeneity.

For each selected study we calculated sensitivity, specificity, the positive Likelihood Ratio (LR+; ratio between the probability of a positive test result given the presence of the disease and the probability of a positive test result given the absence of the disease) and the negative Likelihood Ratio (LR-; ratio between the probability of a negative test result given the presence of the disease and the probability of a negative test result given the absence of the disease). To produce clinically useful pooled statistics, we meta-analyzed LR+ and LR-. For all estimates we added the 95% Confidence Intervals.

3. RESULTS

3.1 Study selection and characteristics

Our search strategy yielded 511 records for consideration (figure 1). After eliminating the 127 duplicates, 384 titles and/or abstracts were reviewed. Of these, 348 were excluded; of the remaining 36 publications, full text-manuscripts were obtained, assessed for eligibility, and a further 24 studies were excluded. Finally, 12 studies were deemed eligible for inclusion and were submitted to data extraction and analysis.¹⁴⁻²⁵ Two

studies reported data separately for Caucasians and Afro-Americans or Caucasians, Afro-American or Asiatic populations from the USA.

The reviewed studies were published between 1998 and 2016; all 12 studies were case-control. Of them, 8 enrolled patients with and without cataplexy, while four studies enrolled only patients with cataplexy. Therefore, 2077 patients with cataplexy (NT1), 235 without cataplexy (NT2) and 7802 ethnically matched controls were included in our meta-analysis. Also, it was possible to analyze 161 patients with hypersomnia/idiopathic hypersomnia. Study populations were similar concerning age [median age: 31 (Inter-Quartile Range IQR 29-32) in patients and 38 (IQR 35-42) years in HC] and gender [median percentage of males: 44 (IQR 38-70) in patients and 54 (IQR: 48-61) in HC].

3.2 Quality of studies

The quality of studies in terms of laboratory methods description, statistical methodology and clinical features was mainly low. The majority of studies did not explain at all or were not clear for the majority of items (Figure 2).

3.3 Meta-analysis of the association between narcolepsy susceptibility and HLA-DQB1 alleles in case-control studies.

All the 12 case-control articles selected for the meta-analysis reported the frequency of the HLA-DQB1*06:02 allele in narcoleptic patients with cataplexy (6 in Asians, 1 in Amerindians, 2 in Afro-Americans, 6 in Caucasians); while 8 works reported the frequency of the HLA-DQB1*06:02 allele in patients without cataplexy (3 in Asians, 1 in Afro-Americans and 5 in Caucasians).^{14, 18, 19, 22}

In the population-stratified analysis, this allele significantly conferred an increased risk for both forms of narcolepsy: NT1 (OR: 24.1, IC: 14.6 – 39.5, $p < 0.001$), as shown in Figure 3, and NT2 (OR: 3.9; IC: 2.2– 6.8, $p < 0.001$), as shown in Figure 4. Since there is evidence of publication bias (Egger test $p = 0.001$ and 0.006), the trim and fill method was used as sensitivity analysis and in both cases the results did not change in a significant way (OR: 22.1, IC: 11.9 – 32.3, $p < 0.001$ NT1, OR: 3.5, IC: 2.3 – 4.9 $p < 0.001$ NT2).

Furthermore, we meta-analyzed the positive and negative LR of developing both forms of narcolepsy if carrying the HLA-DQB1*06:02 allele, calculated from each study.

For narcolepsy with cataplexy, the pooled estimates of LR was intermediate both for positive (5.94, IC 3.71-9.51) and negative LR (0.23, IC 0.16-0.33) (Figure 5), while it was low for narcolepsy without cataplexy (3.35, IC 2.08-5.38 and 0.72, IC 0.63-0.81, respectively) (Figure 6).

3.4 Meta-analysis of the association between subjects with hypersomnia/idiopathic hypersomnia (IH) and HLA-DQB1 alleles in case-control studies.

Four of the 12 case-control articles selected for the main meta-analysis reported the frequency of the HLA-DQB1*06:02 allele in subjects with hypersomnia/idiopathic hypersomnia (1 in Asians, 3 in Caucasians).

In the population-stratified analysis, this allele did not confer an increased risk for hypersomnia (OR: 1.60, IC: 0.6 – 4.30, $p=0.004$), as shown in Supplementary data Figure 1.

Furthermore, we meta-analyzed the positive and negative LR of developing hypersomnia/idiopathic hypersomnia if carrying HLA-DQB1*06:02, calculated from each study. The pooled estimate of LR was low for positive (1.436, IC 0.668- 3.089) and negative LR (0.903, IC 0.714- 1.142).

4. DISCUSSION

The literature indicates that HLA-DQB1*06:02 positivity is the most specific genetic marker linked to narcolepsy, particularly narcolepsy type 1 (NT1), since almost all patients with cataplexy are positive for HLA-DQB1*06:02.⁷ Even though symptom severity does not vary between HLA-DQB1*06:02 homozygous and heterozygous patients, the relative risks increase up to five times in HLA-DQB1*06:02 homozygotes vs. heterozygotes across all ethnic groups studied so far.^{16, 17, 20}

Other HLA alleles or haplotypes have a less striking association with NT1: Tafti and colleagues showed that the HLA-A*11:01, C*04:01, and B*35:01 alleles were associated with susceptibility to NT1 in European NT1 patients that were DQB1*06:02 carriers.²⁶ On the other hand, the Hor et alia study enrolled NT1 and healthy subjects heterozygous for the DRB1*15:01-DQB1*06:02 haplotype and observed a significantly

higher number of DRB1*13:01-DQB1*06:03 carriers among healthy subjects. These data suggest that the DRB1*13:01-DQB1*06:03 haplotype may be a protective factor for NT1.²⁷

One should also consider the presence of other HLA alleles or haplotypes that may be protective or predisposing to NT1. Hor and colleagues enrolled NT1 and healthy subjects heterozygous for the DRB1*15:01-DQB1*06:02 haplotype, and then these authors observed a significantly higher number of DRB1*13:01-DQB1*06:03 carriers among healthy subjects compared to NT1 patients. These data suggest the DRB1*13:01-DQB1*06:03 haplotype may be a protective factor for NT1.²⁷ Another study from Tafti and colleagues showed that in European NT1 patients who were DQB1*06:02 carriers, the HLA-A*11:01, C*04:01, and B*35:01 alleles were also associated with susceptibility to NT1.²⁶

Here we have shown the first meta-analysis reviewing the disease associations of HLA-DQB1*06:02 predisposing to NT1, narcolepsy type 2 (NT2) and hypersomnia. Our study included 12 case-control studies from inception until March 2017 that adhered to all established criteria, evaluating 2077 narcoleptic patients with cataplexy (NT1), 235 narcoleptic patients without cataplexy (NT2), 161 subjects with hypersomnia and 7802 ethnically matched controls for statistical comparisons. No data were available to evaluate the correlation between HLA-DQB1*06:02 and CSF hypocretin-1 levels. We analyzed the frequency of HLA-DQB1*06:02 carriers in subjects from Asia, Europe, North America, and South America.¹⁴⁻²⁵

This meta-analysis unveiled a limitation of the available clinical studies assessing the HLA-DQB1 predisposition to narcolepsy (with and without cataplexy). The quality of studies evaluating the HLA-related genetic background in narcoleptic patients is generally low according to PRISMA standards (as showed in Figure 2).¹² The low methodological quality of studies is surprising, considering that almost all the included articles were published after the first STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines were reported in 2007.²⁸ Moreover, publication bias was observed. Therefore, we had to use the trim and fill method as a sensitivity analysis; in both forms of narcolepsy (with and without cataplexy) the results did not change in a significant way.

While the statistical analyses were usually well described, the methods used to control for confounding factors and subgroups interactions, and the settings of the statistical software programs were either unclear and/or not appropriate in more than half of the published papers. Moreover, most authors never described

any sensitivity analyses, and they never explained how missing data were addressed. In the case-control studies, authors did not clearly explain how matching of cases and controls was done, and if a consecutive or random sample of patients were enrolled. However, although more than half of articles do not consider Hardy-Weinberg equilibrium and do not define their methods for inferring genotypes/haplotypes, the analytical and genotyping methods were clearly defined in almost all of the selected studies. This observation reassures us regarding the reliability of the immunogenetics data analyzed.

To produce clinically useful pooled statistics, we did a meta-analysis of positive and negative Likelihood Ratio (LR+ and LR-) to develop NT1, NT2, and hypersomnia, if carrying HLA-DQB1*06:02 (Figure 5, Figure 6, Supplementary data Figure 1).

We confirmed that HLA-DQB1*06:02 positivity supports the diagnosis of NT1. For NT1, LR+ was 5.94 (IC 3.71-9.51) meaning a clinically relevant increase of risk, and LR- was 0.23 (IC 0.16-0.33), thus suggesting a five-fold risk of being narcoleptic with cataplexy among HLA-DQB1*06:02 carriers and 0.3-fold risk to be narcoleptic among non-carriers. Moreover, the risk stratification in our meta-analysis confirmed the risk of positivity for the HLA-DQB1*06:02 allele, regardless of ethnicity. Furthermore, HLA-DQB1*06:02 negativity suggests that the patient does not suffer from NT1 and that the subject is not hypocretin-deficient. For NT2, the pooled estimate of LR+ was 3.35, meaning a clinically useful increase of risk, and LR- was 0.72 (IC 0.63-0.81), also suggesting an association between being a carrier of HLA-DQB1*06:02 and developing NT2, although to a slightly lesser degree with respect to NT1. Thus, clinicians should not exclude the possibility of NT2 in HLA-DQB1*06:02 carriers.

Finally, for hypersomniacs (IH) LR+ was 1.436 (IC 0.668- 3.089) and LR- was 0.903 (IC 0.714- 1.142), suggesting no relationship between hypersomnia/idiopathic hypersomnia and HLA-DQB1*06:02.

Our meta-analysis represents an evidence-based confirmation of the role of the HLA-DQB1*06:02 allele in susceptibility to narcolepsy type 1, and an indication of increased vulnerability for narcolepsy type 2. An increased use of HLA-DQB1 typing should be encouraged for practical purposes in the diagnostic process of NT1 and NT2.

The presence of HLA-DQB1*06:02, even though strongly associated with narcolepsy, is neither

sufficient nor necessary for developing NT1 and has limited diagnostic value. As very few HLA-DQB1*06:02 negative subjects displaying normal hypocretin levels with narcolepsy and cataplexy have been reported, HLA typing should always be considered within the global clinical picture of each subject.²⁹

In NT2, an HLA-DQB1*06:02 association should be considered with caution in a clinical setting since NT2 is still a poorly defined condition and consists of a heterogeneous phenotype.³⁰ Moreover, given the difficulties regarding hypocretin measurements and recognition of cataplexy, subjects diagnosed with narcolepsy without cataplexy (ICSD-2nd) may have deficits of hypocretinergic neurotransmission. In the same vein, subjects with a diagnosis of NT2 according to ICSD-3rd criteria may have a hypocretin deficiency (if not measured) or a hypocretin deficiency could not be found due to current radioimmunoassay's lack of precision. Patients with NT2 can develop cataplexy after daytime somnolence, forcing a change in diagnosis to NT1. Indeed, it must be kept in mind that there is a constant risk of misdiagnosing NT2 instead of NT1 in subjects who do not undergo hypocretin level determination. Hypocretin-deficient subjects after lumbar puncture must be reclassified as NT1. On these criteria, clinicians must consider that HLA DQB1*06:02 positivity in NT2 subjects without hypocretin level determination, involves a risk of misdiagnosing NT2 as NT1. On the other hand, papers classifying NT2 subjects without lumbar puncture carry a risk of magnifying the frequency of DQB1*06:02 positivity in NT2.

All in all, HLA typing could be used as a complementary diagnostic marker that has the potential to improve sensitivity for diagnosing different types of hypersomnia (NT1, NT2, and IH).²⁹

Since IH has no relationship with HLA DQB1*06:02, HLA typing is a valuable and easy-to-handle tool to support the diagnosis of NT1/NT2. HLA DQB1*06:02 positivity in NT2, from a clinical point of view, should encourage measurement of hypocretin, potentially underlining a hypocretin deficit.

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FIGURE LEGENDS

Figure 1. Flow Diagram of the study based on the PRISMA statement.

Figure 2. Analysis of the quality of the studies included in the meta-analysis.

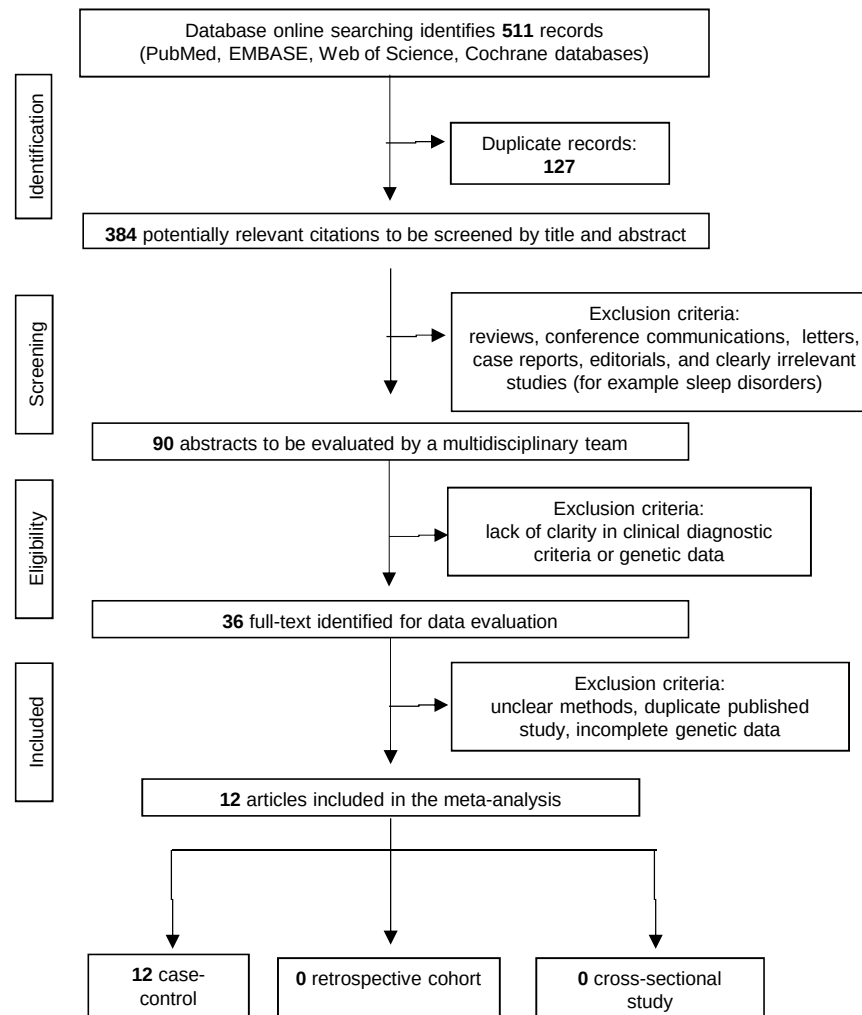
Figure 3. The risk for NT1 patients carrying the HLA-DQB1*06:02 allele. OR= Odds Ratio. CI= Confidence Interval.

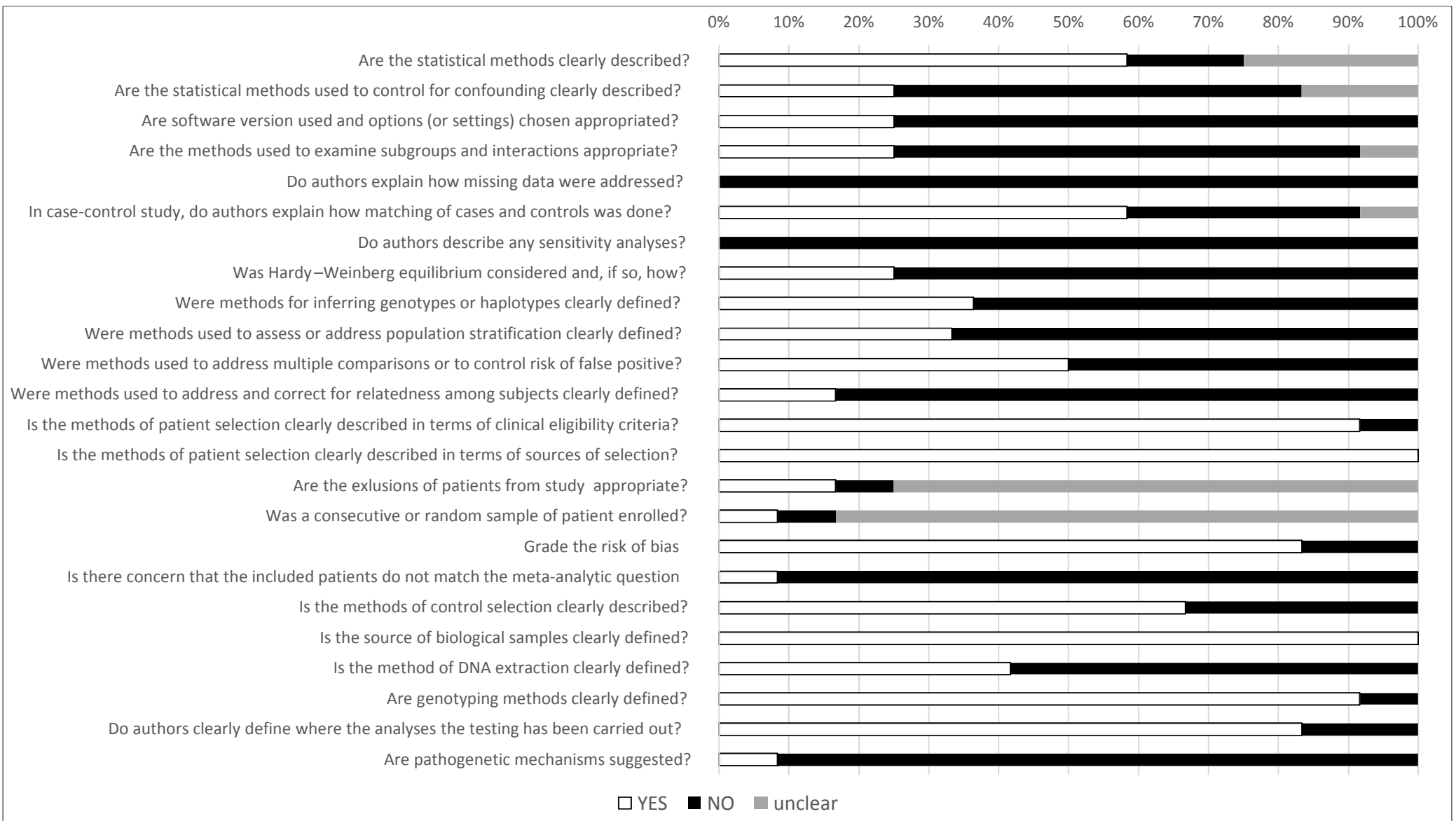
Figure 4. The risk for NT2 patients carrying the HLA-DQB1*06:02 allele. OR= Odds Ratio. CI= Confidence Interval.

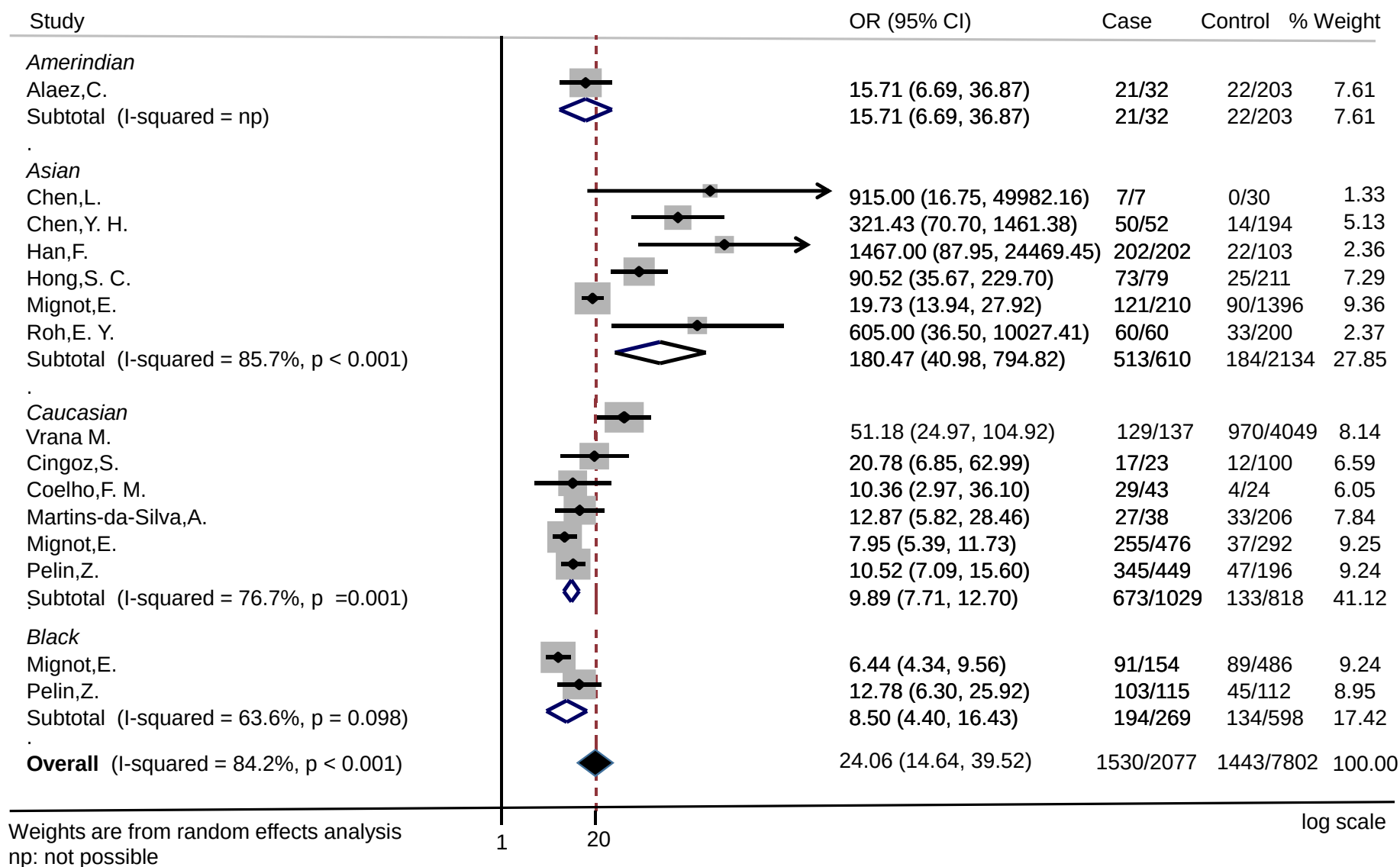
Figure 5. Positive (top) and negative (bottom) Likelihood Ratio (LR) to develop NT1 carrying the HLA-DQB1*06:02 allele. CI= Confidence Interval.

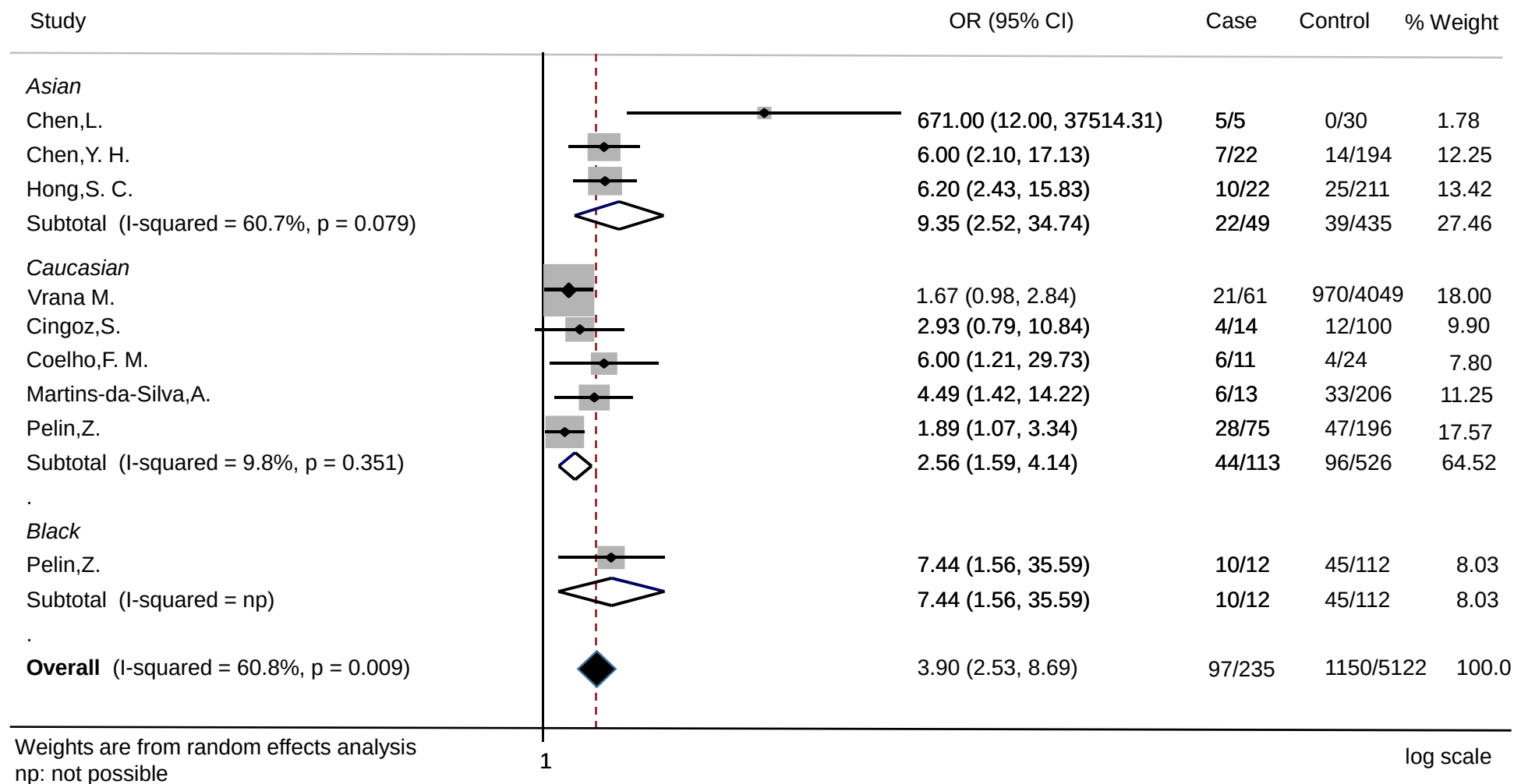
Figure 6. Positive (top) and negative (bottom) Likelihood Ratio (LR) to develop NT2 carrying the HLA-DQB1*06:02 allele. CI= Confidence Interval.

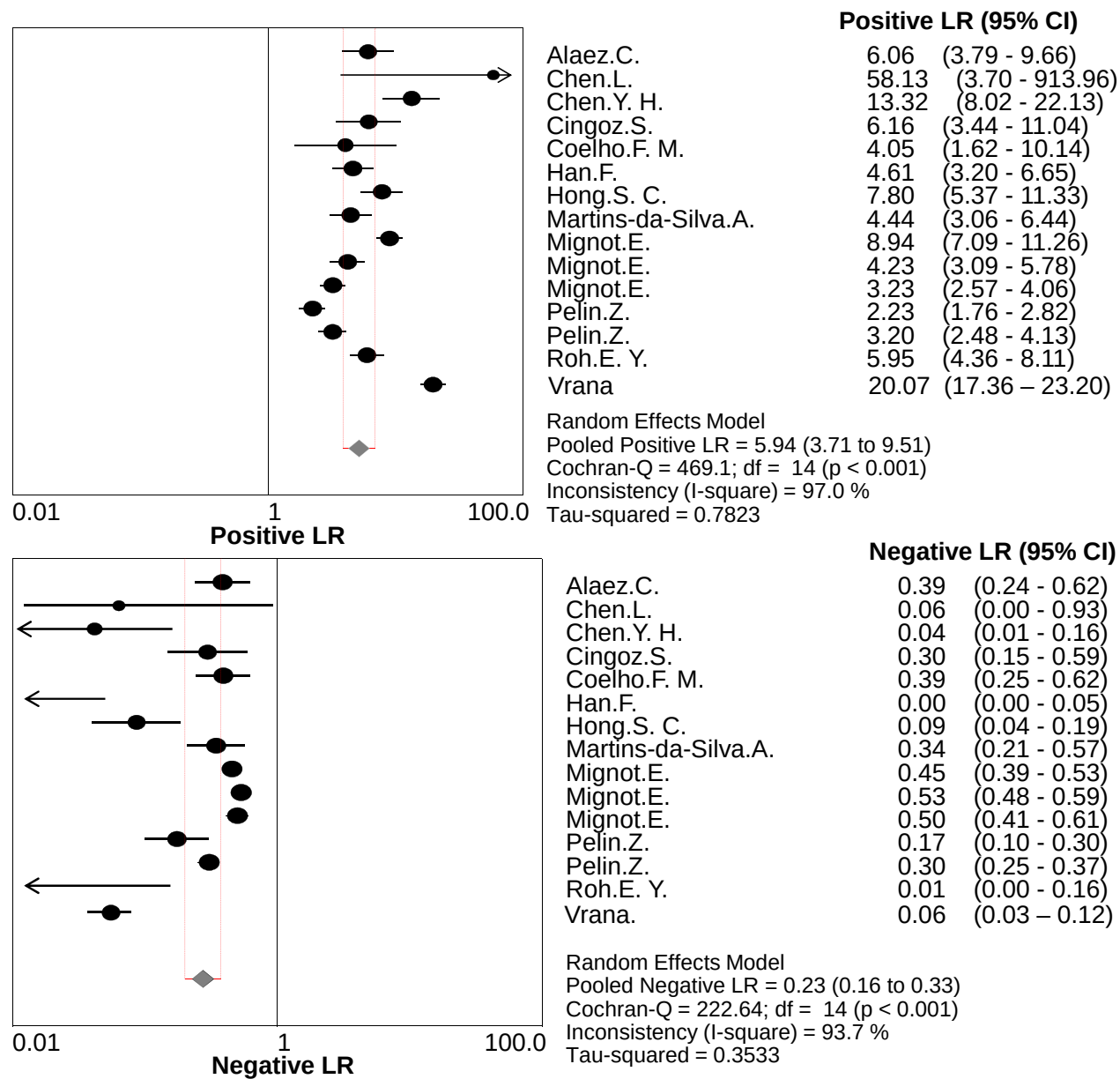
Figure 1 Supplementary data. The risk for patients with hypersomnia/idiopathic hypersomnia carrying the HLA-DQB1*06:02 allele. OR= Odds Ratio. CI= Confidence Interval.

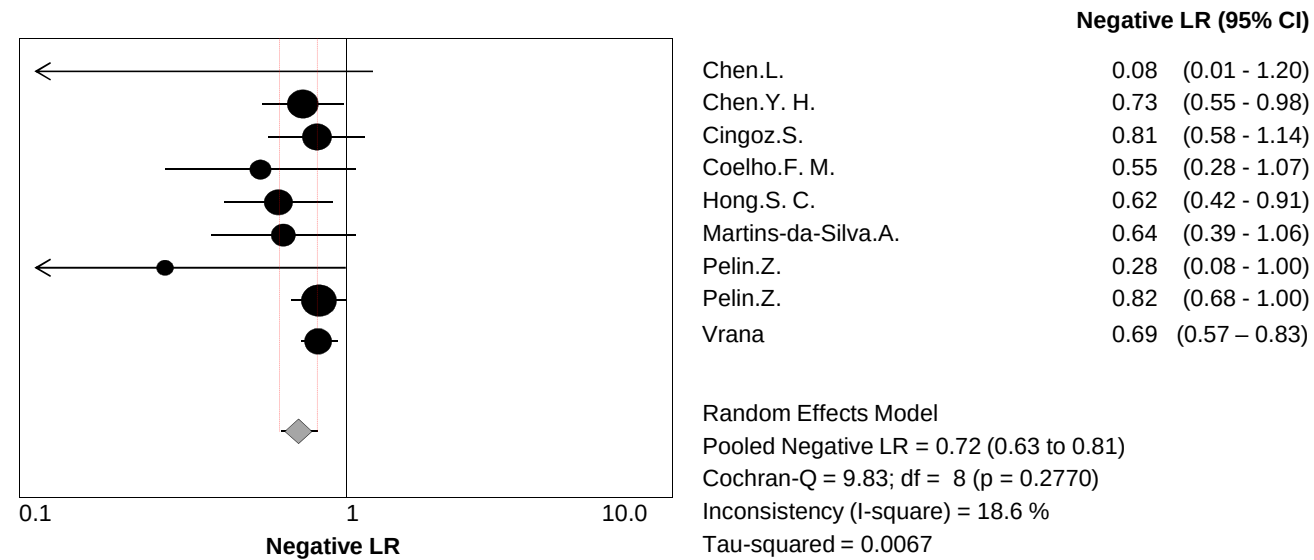
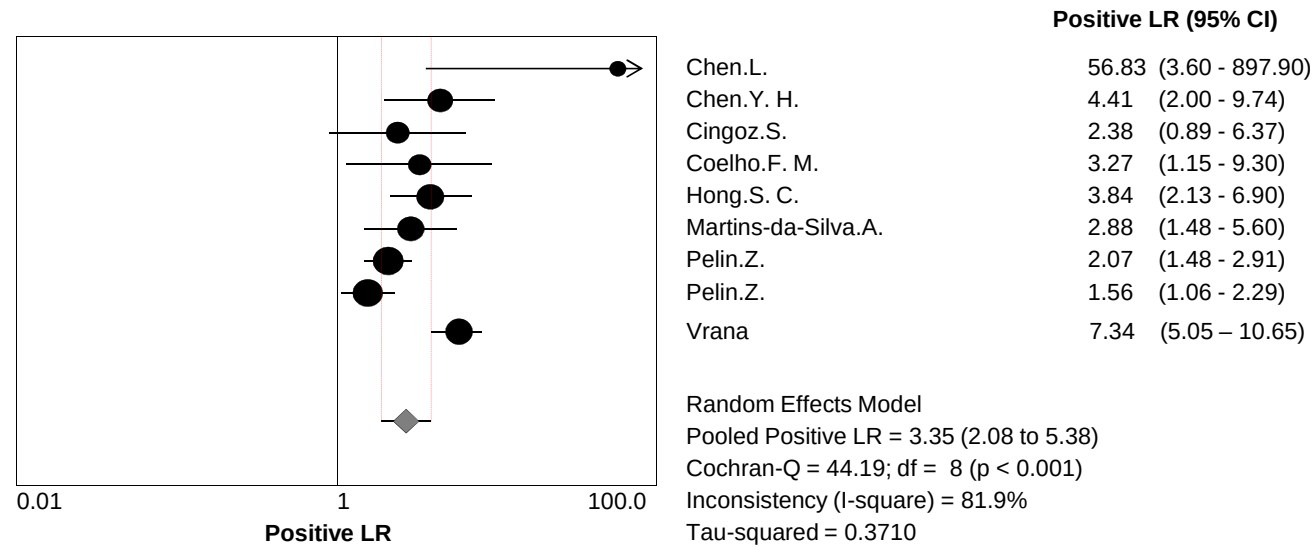












Highlights

- HLA-DQB1*06:02 conferred an increased risk for NT1 and, to a lesser extent, NT2
- No relationship between hypersomnia/idiopathic hypersomnia and HLA-DQB1*06:02 was found
- High risk carried by HLA-DQB1*06:02 allele is present in all ethnic groups
- HLA-DQB1 typing is still useful in the diagnosis of narcolepsy with cataplexy.